

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit III | Session 2 | SLO 2: Inter-Thread Communication (Code: 3022)

Q: 1. Fill in the blank:

The wait(), notify(), and notifyAll() methods belong to the _____ class.

Ans: Answer: Object

Q: 2. True or False:

wait() releases the monitor lock while the thread is waiting.

Ans: Answer: True

Q: 3. Multiple Choice:

Which method wakes up all threads waiting on an object's monitor?

A) notify() B) resume() C) notifyAll() D) wakeAll()

Ans: Answer: C) notifyAll()

Q: 4. Fill in the blank:

wait() must always be called inside a _____ context.

Ans: Answer: synchronized

Q: 5. What exception is thrown if wait() is called without holding the monitor lock?

Ans: Answer: IllegalMonitorStateException

Q: 6. True or False:

Thread.sleep() releases locks while sleeping.

Ans: Answer: False (sleep() holds locks, whereas wait() releases the lock).

Q: 7. Fill in the blank:

To guard against spurious wakeups, wait() should always be placed inside a _____ loop.

Ans: Answer: while

Q: 8. Multiple Choice:

Which classical synchronization problem demonstrates inter-thread communication?

A) Producer-Consumer B) QuickSort C) Binary Search D) Bubble Sort

Ans: Answer: A) Producer-Consumer

Q: 9. Identify the bug: Calling notify() outside synchronized block.

Ans: Answer: Results in IllegalMonitorStateException at runtime.

Q: 10. Fill in the blank:

notify() wakes up an _____ thread waiting on the object monitor.

Ans: Answer: arbitrarily chosen (single)